Lecture 9: Muscle Engineering

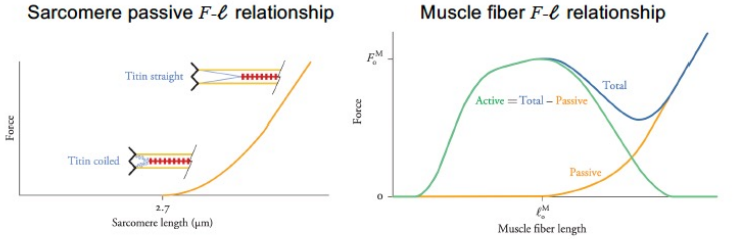
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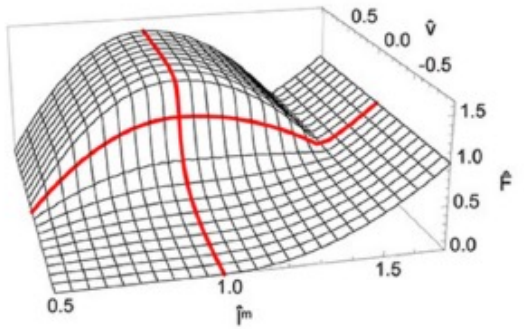
Review: Muscle biomechanics

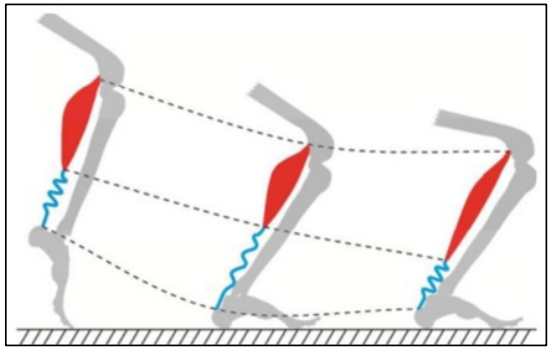
Force-length relationship (active + passive)



F_0^M peak active force at ℓ_0^M optimum muscle fiber length

Active+passive F-ℓ-v relationship at full activation



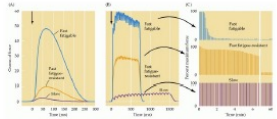


Fiber Types: myosin heavy chain expression

- Type I: Slow-twitch oxidative (SO)
  - Produce ATP aerobically
  - Rich in capillaries, mitochondria and myoglobin (red color)
  - Slow contraction time (>70 ms)
  - Relatively small diameter, relatively little tension
  - Slow to fatigue
- Type IIa: Fast-twitch oxidative (FO)
  - Produce ATP aerobically
  - Rich in mitochondria but not in myoglobin (lighter than SO)
  - Relatively fast contraction (30–50 ms)
  - Will eventually fatigue
- Type IIx: Fast-twitch glycolytic (FG)
  - Produce ATP via anaerobic glycolysis
  - Few mitochondria and little myoglobin – (light/white color)
  - Fast contraction (30–40 ms)
  - Usually large diameter, produce great tension for a short time
  - Fast to fatigue
- (Type IIb: not in humans. Largest, fastest and quickest to fatigue)
- Average population: 50-55% I, 30-35% IIA, 15% IIx. Proportion SO increases with age.
- Varies greatly among muscles & individuals (SO in Quads: 20% in sprinters -> 95% in marathon runners)



Talbot & Maves, 2017



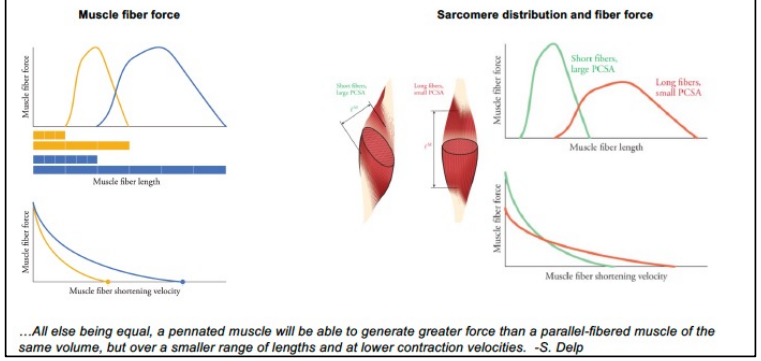
Motor unit size

The number of muscle fibers in a motor unit varies among muscles.

- In the hand and eye, fewer than **100 muscle fibers** can occupy a motor unit.
- In the lower leg, a single motor unit may contain as many as **1,000 muscle fibers**

Muscles which possess more motor units (and thus have greater individual motor neuron innervation) are able to control force output more finely.

Sarcomere organization and fiber force



…All else being equal, a pennated muscle will be able to generate greater force than a parallel-fibered muscle of the same volume, but over a smaller range of lengths and at lower contraction velocities. -S. Delp

Discussion: Muscle-tendon unit engineering

Design muscle-tendon units for:

Finger flexor	**Brain surgeon**	**Rock climber**
Property A		
Property B….		
….		

Ankle plantarflexor	**High jumper**	**Marathon runner**
Property …		

Knee extensor	**Cyclist**	**Ballet dancer**
Property …		

(open question)

Properties can include but are not limited to…….

- Diameter (S-M-L)
- Pennation (non-, uni-, bi-/multi-)
- Fiber type composition (sum to 100%)
- Relative optimal fiber length (short, med, long)
- Tendon slack length (short, med, long) and stiffness (stiff or compliant)
- Motor unit size

- Anything else? Different attachment points? Rerouting? Different joint type?